

Homework 12:

100/100

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Exam 3 problem 1:

(50/50)

liquid

Solid $y=L$

$y=0$
 $C_A = C_0$

$$R_{AB} = -k_B$$

$$R_{AC} = -k_C$$

$$R_{AF} = -k_F C_A$$

0th order $A \rightarrow B$

0th order $A \rightarrow C$

1st order $A \rightarrow F$

$$C_A(0) = C_0$$

1st order: $D \frac{d^2 C_A}{dy^2} - k_F C_A = 0$

$$y = \sqrt{\frac{k_F}{D}}$$

$$\frac{d}{dy} \left(D \frac{dC_A}{dy} \right) - k_B - k_C - k_F C_A = 0$$

$$B.C.1 \frac{dC_A}{dy} = 0 \text{ @ } y=L$$

$$B.C.2 \text{ @ } y=0, C_A(0) = C_0$$

$$0^{th} \text{ order: } C_A(y) = C_1 e^{\sqrt{\frac{k_F}{D}} y} + C_2 e^{-\sqrt{\frac{k_F}{D}} y} - \frac{k_B + k_C}{k_F}$$

$$C_A(0) = C_1 + C_2 - \frac{k_B + k_C}{k_F} = C_0$$

$$\frac{dC_A}{dy} = \sqrt{\frac{k_F}{D}} C_1 e^{\sqrt{\frac{k_F}{D}} y} - \sqrt{\frac{k_F}{D}} C_2 e^{-\sqrt{\frac{k_F}{D}} y} = 0$$

solve for C_1

$$C_1 = C_0 - C_2 + \frac{k_B + k_C}{k_F}$$

$$C_0 e^{\sqrt{\frac{k_F}{D}} L} - C_2 \left(e^{\sqrt{\frac{k_F}{D}} L} + e^{-\sqrt{\frac{k_F}{D}} L} \right) + \frac{k_B + k_C}{k_F} e^{\sqrt{\frac{k_F}{D}} L} = 0$$

→ solve for C_2

$$C_2 = \frac{C_0 e^{\sqrt{\frac{k_F}{D}} L} + \frac{k_B + k_C}{k_F} e^{\sqrt{\frac{k_F}{D}} L}}{e^{\sqrt{\frac{k_F}{D}} L} + e^{-\sqrt{\frac{k_F}{D}} L}}$$

plug back into C_1

$$C_1 = \frac{C_0 - e^{\sqrt{\frac{k_F}{D}} L} \left[\frac{C_0 e^{\sqrt{\frac{k_F}{D}} L} + \frac{k_B + k_C}{k_F} e^{\sqrt{\frac{k_F}{D}} L} \right] + \frac{k_B + k_C}{k_F}}{e^{\sqrt{\frac{k_F}{D}} L} + e^{-\sqrt{\frac{k_F}{D}} L}}$$

$$C_A(y) = \left(\frac{C_0 - e^{\sqrt{\frac{k_F}{D}} L} \left(C_0 + \frac{k_B + k_C}{k_F} \right) + \frac{k_B + k_C}{k_F}}{e^{\sqrt{\frac{k_F}{D}} L} + e^{-\sqrt{\frac{k_F}{D}} L}} \right) e^{\sqrt{\frac{k_F}{D}} y} + \left(\frac{e^{\sqrt{\frac{k_F}{D}} L} \left(C_0 + \frac{k_B + k_C}{k_F} \right) - \frac{k_B + k_C}{k_F}}{e^{\sqrt{\frac{k_F}{D}} L} + e^{-\sqrt{\frac{k_F}{D}} L}} \right) e^{-\sqrt{\frac{k_F}{D}} y} - \frac{k_B + k_C}{k_F}$$

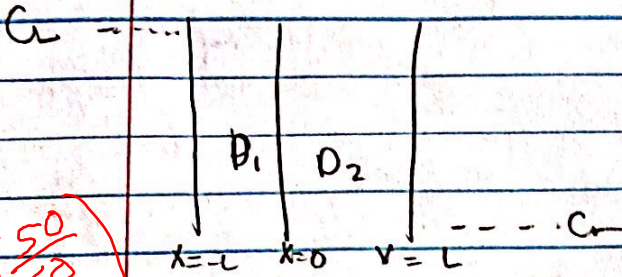
$$N_A = -D_A \frac{dC_A}{dy}$$

$$N_A = -D_A \frac{dC_A}{dy} = -D_A \left(\sqrt{\frac{k_F}{D}} \left(\frac{C_0 - e^{\sqrt{\frac{k_F}{D}} L} \left(C_0 + \frac{k_B + k_C}{k_F} \right) + \frac{k_B + k_C}{k_F}}{e^{\sqrt{\frac{k_F}{D}} L} + e^{-\sqrt{\frac{k_F}{D}} L}} \right) e^{\sqrt{\frac{k_F}{D}} y} - \sqrt{\frac{k_F}{D}} \left(\frac{e^{\sqrt{\frac{k_F}{D}} L} \left(C_0 + \frac{k_B + k_C}{k_F} \right) - \frac{k_B + k_C}{k_F}}{e^{\sqrt{\frac{k_F}{D}} L} + e^{-\sqrt{\frac{k_F}{D}} L}} \right) e^{-\sqrt{\frac{k_F}{D}} y} \right)$$

Bonus:

$$R_A = -k_B - k_C - k_F C_A$$

Problem 2:



M1: $D_1 \frac{d^2 C_1}{dx^2} = 0 \xrightarrow{\text{integrate}} Ax + B = C_1(x)$

M2: $D_2 \frac{d^2 C_2}{dx^2} - K C_2 = 0 \xrightarrow{\text{integrate}} -\frac{K_0 L^2}{2D_2} + Ex + F = C_2(x)$

B.C. 1) $C_1(x=-L) = C_L$

B.C. 2) $C_2(x=0) = C_L$

B.C. 3) @ $x=0$ $\phi = \frac{C_2(x=0)}{C_1(x=0)} = 1$

B.C. 4) $-D_1 \frac{dC_1}{dx} \big|_{x=0} = -D_2 \frac{dC_2}{dx} \big|_{x=0}$

$-AL + B = C_L$

$\frac{EL}{D_1} + F\phi = C_L$

$-\frac{K_0 L^2}{2D_2} + EL + F = C_L$

$\frac{EL}{D_1} + [C_L - EL + \frac{K_0 L^2}{2D_2}] \phi = C_L$

$\phi = \frac{B}{F} \Rightarrow B = F\phi$

$E[\frac{L}{D_1} - \phi] = C_L - [\frac{K_0 L^2}{2D_2} + C_L] \phi$

$-D_1 A = E$

$E = C_L - [C_L + \frac{K_0 L^2}{2D_2}] \phi$

$A = -\frac{E}{D_1}$

$\frac{L}{D_1} - L\phi$

$A = -C_L - [C_L + \frac{K_0 L^2}{2D_2}] \phi$

$B = C_L - C_L [C_L + \frac{K_0 L^2}{2D_2}] \phi$

$L - L D_1 \phi$

$1 - D_1 \phi$

$F = C_L + \frac{K_0 L^2}{2D_2} - \frac{C_L - [C_L + \frac{K_0 L^2}{2D_2}] \phi}{\frac{L}{D_1} - L\phi}$

$\frac{1}{D_1} - \phi$

$C_1(x) = \frac{-C_L - [C_L + \frac{K_0 L^2}{2D_2}] \phi}{L - D_1 \phi L} x + \frac{C_L - C_L [C_L + \frac{K_0 L^2}{2D_2}] \phi}{1 - D_1 \phi}$

$C_2(x) = \frac{-K_0 x^2}{2D_2} + \frac{C_L - [C_L + \frac{K_0 L^2}{2D_2}] \phi}{\frac{L}{D_1} - L\phi} x + C_L + \frac{K_0 L^2}{2D_2} - \frac{C_L - [C_L + \frac{K_0 L^2}{2D_2}] \phi}{\frac{1}{D_1} - \phi}$

Bonus: $N_{A1} = D_1 [-C_L - [C_L + \frac{K_0 L^2}{2D_2}] \phi] / (L - D_1 \phi L)$

$N_{A2} = -K_0 x + D_2 \left[\frac{C_L - [C_L + \frac{K_0 L^2}{2D_2}] \phi}{\frac{L}{D_1} - L\phi} \right]$